

PART 1 – Emotional Interaction (Chapter 6)

1. Why emotions matter in HCI now

Old HCI:

“Make the system efficient and effective. Fast. Fewer clicks.”

New HCI:

“How can we make people **feel** a certain way when they use technology?”

- Maybe we want them to be:
 - **Happy** (entertainment, games)
 - **Trusting** (banking, health apps)
 - **Motivated** (fitness, study apps)
 - **Comforted / less lonely** (social robots, chatbots)

Emotional interaction = how we **feel and react** during interaction with technology.

Affective computing uses recognition software + machine learning to **sense and respond** to emotions.

2. What emotional interaction looks at

Key questions in your slides:

- What makes us happy, sad, annoyed, anxious, motivated, etc.?
- Why do people get emotionally attached to products (like **virtual pets**)?
- Can social robots help **reduce loneliness** and improve well-being?
- Can we **change human behavior** through emotional feedback?

So emotional interaction isn't just “nice UI”, it's also:

- **Persuasion** (make you walk more, save energy)
 - **Companionship** (robots in care homes)
 - **Nudging** behavior (Amazon 1-click, reminder apps)
-

3. Tiny emotional tricks: smiley buttons & sad ads

3.1 Emoji feedback box (Heathrow example)

Slide: “How was your security experience today?” → 3 big smiley buttons (green, yellow, red).

Why is it effective?

- **Effortless** – just tap a face. No typing.
- **Immediate emotional mapping** – you don’t think, just press your feeling.
- Works even if:
 - You’re tired
 - You don’t know English
- It’s **fun**, so people actually use it.

3.2 “Pulling at the heart strings” – homeless donation ad

Slide shows a homeless person and a message asking to help them “take their first step out of homelessness.”

- Uses **emotive imagery** + text to trigger:
 - **Empathy**
 - **Guilt**
 - **Desire to help**
- Emotion → action → **donation**.

4. Can we design for moods?

Slide literally asks: “Is it possible to design an interface to match or change how we are feeling?”

Questions it raises:

- Should interfaces try to **improve our mood**? If yes, **how**?
- Our mood constantly changes; how does the system:
 - Track it?
 - Decide when to respond?
- What kind of interface suits:
 - **Happy** person? Maybe bright, playful, humorous
 - **Sad** user? Gentle, simple, low cognitive load
 - **Angry** user? Minimal friction, clear feedback
 - **Bored** user? More exploration, discovery
 - **Focused** user? No distractions

There's **no fixed formula**, but the point is:

Interface design must consider **emotion & context**, not just functionality.

5. Emotions ↔ Behavior

Slide questions:

- When you're **angry**, do you concentrate better or get distracted?
- When you're **happy**, do you take more risks (spend more)?

- Baumeister et al. (2007): the relationship isn't a simple "one emotion = one behavior" formula. It's more complex.

So emotional state:

- Changes how we:
 - Pay attention
 - Take risks
 - Make decisions
-

6. Automatic vs conscious emotions

Emotions can be:

- **Short-lived** (a flash of anger)
- **Long-lasting** (jealousy, resentment)

Two categories:

1. Automatic emotions

- **Fast**, triggered immediately.
- Fade quickly.
- Example: your laptop freezes → you instantly feel annoyed.

2. Conscious emotions

- Develop **slowly**.
- Last longer (you keep thinking about it).
- Example: long-term bitterness about being treated unfairly.

Design implication:

- Automatic emotions are targeted by **instant feedback** (e.g., red error border, success tick).
 - Conscious emotions may be targeted by **branding**, long-term experience, trust-building.
-

7. Ortony / Norman-style three levels of emotional design

Slide shows the model: **Visceral – Behavioral – Reflective**.

7.1 Claims from the model

- When **frightened / angry**:
 - Focus narrowly
 - Body tenses, you sweat
 - You become **less tolerant** of problems.
- When **happy**:
 - Less focused, more relaxed
 - More willing to overlook small problems
 - More **creative**.

So: mood changes **how forgiving** we are to UI flaws.

7.2 Three design levels (MEMORIZE)

1. Visceral design

- Looks, feel, sound: immediate first impression.
- “Is this pretty? Does it feel cool?”

2. Behavioral design

- Actual **use**: usability, performance, function.
- “Is it easy to use? Does it work?”

3. Reflective design

- Meaning & personal value.
- “What does owning this say about me? Do I feel proud?”

👉 **MEMORIZE**: Visceral = look/feel, Behavioral = usability, Reflective = meaning.

7.3 Swatch watch example

- **Visceral** – bright colors, wild design attract attention.
 - **Behavioral** – shape, strap, and buttons make it easy to use as a watch.
 - **Reflective** – cultural images/graphics reflect style, identity, brand.
-

8. Plutchik’s wheel of emotions

Slide shows a flower-like wheel.

- Categorizes 7 “core” emotions:
 - **Anger, Disgust, Fear, Sadness, Anticipation, Joy, Surprise**
 - Plus **Trust** (not usually considered an emotion).
- Adds labels like:
 - Optimism, love, submission, awe, disapproval, remorse, contempt, aggressiveness.
- **Color intensity** indicates strength:
 - Darker → more intense.
- Designers can use it like a **color palette / mood board** to elicit certain feelings.

- BUT: it **does not tell you exactly how to design** for each emotion; it's more descriptive.
-

9. Expressive interfaces

An **expressive interface**:

- Uses **color, icons, sounds, animations** to convey emotional tone.
- Provides **reassuring feedback** that can be informative AND fun.
- But can also be **intrusive** or annoying.

Key point:

People sometimes tolerate slow or imperfect systems if the interface is **pleasing and aesthetic**.

Examples from slides:

- Old Mac “smiling face” vs modern spinning beachball for “please wait”.
 - Nest thermostat:
 - Round, clean, large type → friendly, minimalist
 - Compared to old ugly box thermostats.
 - Aesthetic clothing website hero image.
-

10. Affective computing & Emotional AI

Affective computing:

Use computers to **recognize and express emotions** like humans.

- Uses sensors:
 - **GSR** (sweat on skin)

- **Facial expressions** (camera)
- **Gestures, body movement** (accelerometers, motion capture)
- Explores how affect (emotion) affects **personal health**.

Emotional AI:

- Uses AI to **infer feelings** from:
 - Face
 - Voice
 - Text (tweets, posts)
- Goal: predict:
 - What you're likely to **buy** when you're bored / sad / happy.
 - How you behave (e.g., risk, driving style).

Techniques:

- Cameras for facial expressions
- Biosensors for GSR
- Voice tone (intonation, pitch, loudness)
- Body movement sensors

Classification of emotions:

- Measures 6 core expressions:
 - Sadness, disgust, fear, anger, contempt, joy.
- Detects features like:

- Smiling, eye widening, brow raise/furrow, cheek raise, mouth open, upper-lip raise, nose wrinkle.

Affdex example image: phone analyzing face in real time.

11. How this emotional data is used (and ethics)

If system detects:

- **Disgust** when an ad shows → may change ad.
- **Happiness** while watching → choose similar content.

More examples:

- In a car: if driver looks **angry**, system might suggest a break.
- Eye-tracking, pulse, speech, and social media text also used.

Indirect emotion detection:

- Predict if someone:
 - Is suitable for a certain job
 - May vote a certain way
- Big ethical question:

Is it okay for technology to “read” your emotions and use them commercially?

12. Detecting & reflecting on moods

Moods are more subtle than emotions and harder to detect automatically.

- We can be happy one day, low the next.
- Mood is often **not strongly visible** in physiological data.

So tech helps by letting users **self-track**:

- **Mood tracker apps** (Moodnotes, Daylio)
 - Users log mood over time
 - Reflect why they feel gloomy or cheerful
 - Reflection → assumed to improve well-being.
 - **VR Mood Worlds**
 - People explore their moods in a VR environment.
-

13. Behavior change & persuasive tech

Systems designed to change people's **attitudes and behaviors**.

Examples of techniques:

- Pop-up ads
- Warning messages
- Reminders, prompts
- Personalized messages
- Recommendations
- Amazon 1-click → encourages quick purchasing

Collectively called **nudging**.

13.1 Pocket Pikachu

- A small device with a digital pet that rewards **physical activity**.
- You have to walk/run/jump to keep Pikachu happy.
- If you don't, it becomes **angry and refuses to play**.

Behavior change logic:

- **Emotional attachment** → you want Pikachu to be happy.
- If it sulks, you feel bad → motivates you to walk.

13.2 Voice assistants & manners

- Kids talk to Alexa casually, often without “please” and “thank you”.
- Question: will they copy this in real life with humans?
- Should Alexa enforce etiquette (e.g., “try asking more politely”)?
- Research: children **do** treat humans differently from voice assistants.

13.3 Tracking devices

- Fitness apps, sleep trackers, calorie counters.
- Compare your stats with friends on leaderboards.
- Apps encouraging reflection → higher happiness.

13.4 Sustainable HCI & Tidy Street

- Design tech to help reduce **energy use**.
- Feedback (infographics, emoticons) encourages reflection.
- **Tidy Street project**:
 - Chalk visualization of a street's energy use on the road.
 - Real-time feedback everyone can see → **15% reduction** in consumption.

13.5 Phishing

- Using web/email to **deceive** people (PayPal, lottery scams).
 - Internet boosts the **scale and creativity** of deception.
-

14. Anthropomorphism (MEMORIZE definition)

Anthropomorphism = giving human-like qualities to non-human things (objects, computers, robots).

- Used heavily in ads (dancing butter, cereal characters).
- Used in HCI to:
 - Make UX enjoyable
 - Reduce anxiety
 - Create personalities people like to interact with.

Examples:

- Friendly welcome message vs “User 24, commence Exercise 5.”
- “Now Chris, that’s not right... Try again” vs “Incorrect”.

14.1 Classic HCI examples: Clippy & Anna

- **Clippy**: animated paperclip helper in MS Office.
 - Intended to help, but users found it:
 - Annoying, patronizing, distracting.
- **IKEA’s Anna**:
 - Virtual agent with blinking, lip and head movements to appear more human.

14.2 Error messages

Bad example:

“The application Word Wonder has unexpectedly quit due to a type 2 error.”

Better:

“The application has expectedly quit due to poor coding...” (joke)

Shneiderman's guidelines for error messages:

- Avoid words like **FATAL, INVALID, BAD**
- Careful with **audio warnings** (startle).
- Avoid ALL CAPS + long code numbers.
- Be **precise**, not vague.
- Provide **context-sensitive help**.

Fun 404 page example with Lego dude.

14.3 Evidence & dilemmas

- Reeves & Nass (1996):
 - Computers that **flatter/praise** users (“Great job!”) → positive impact, students continued more.
- They argue computers should **apologize**:
 - “I’m really sorry I crashed. I’ll try not to do it again.”
- Debate:
 - Would users find it **sincere** or fake?
 - How should computers show politeness?

14.4 Social robots & attachment

- Robots like **Paro** (seal robot), **Stevie** in care homes.

- Help with:
 - Games, jokes, singalongs
 - Social interaction
 - Some residents love them; some feel it's demeaning or weird.
 - Conclusion slide:
 - Robots can play **supporting** role, but they can never fully replace **human warmth**.
-

15. Emotional Interaction – Summary to MEMORIZE

From the last slide:

- Emotional aspects deal with facilitating **positive states** (pleasure) and avoiding **negative reactions** (frustration).
 - Well-designed, aesthetic interfaces can be a **pleasure to use**.
 - Emotional AI + affective computing detect emotions from **faces, conversations, sensors**.
 - Emotional tech can be used to **persuade behavior change** (fitness, energy, etc.).
 - Anthropomorphism = human qualities to objects.
 - Robots are increasingly used as **companions**.
-

PART 2 – Basic Design Rules (Influencing Perception)

File: “Basic Design Rules – HCI Week 8”

Themes:

How can I influence **how a design is perceived**?

Rules in your slides:

- Accessibility
 - 80/20 Rule
 - Affordance (+ mental model)
 - Hick's Law
 - Fitts' Law
 - Golden Ratio
 - Fibonacci Sequence
 - Rule of Thirds
 - Signal-to-Noise Ratio
-

1. Accessibility (MEMORIZE 4 characteristics)

Definition:

Designs should be usable by **people of diverse abilities**, without special adaptation.

Related terms: **barrier-free design, universal design, inclusive design.**

Four characteristics:

1. **Perceptibility**

- Everyone can **perceive** the design, regardless of sensory ability.
- Example: good contrast, captions for audio.

2. **Operability**

- Everyone can **use** it, regardless of physical ability.
- Example: keyboard navigation, large tap targets.

3. **Simplicity**

- Everyone can **understand and use** it easily, regardless of experience, literacy, focus level.
- Example: clear language, simple flows.

4. **Forgiveness**

- Minimize **errors** and their **consequences**.
- Example: undo, confirmation dialogs, autosave.

👉 **MEMORIZE:** Perceptibility, Operability, Simplicity, Forgiveness.

2. **80/20 Rule (Pareto principle)**

Definition:

~80% of the effects come from **20%** of the causes/variables.

Examples from slide:

- 80% of product usage = 20% of features
- 80% of traffic = 20% of roads
- 80% of revenue = 20% of products
- 80% of innovation = 20% of people
- 80% of progress = 20% of effort
- 80% of errors = 20% of components

Design implications:

- Focus on the **vital few** features that people use the most.
- Non-critical functions in the less-important 80% can be:

- De-prioritized
 - Hidden
 - Or eliminated to reduce complexity.
-

3. Affordance & Mental Models

Affordance:

Physical properties of an object/environment **suggest** what actions are possible.

- Good affordance:
 - Door handle that **looks pullable** when you should pull.
- Bad affordance:
 - Flat metal plate on a door that actually needs to be pulled (confusing).

Design guidance:

- Design objects to **afford their intended function**.
- Also **negatively afford** improper use (make wrong use harder).

Mental Models:

- Users build an internal story of “how the system works” (even if it’s wrong).
- To match mental models, we often **hide implementation details**.
- Design should be:
 - Simple
 - Familiar
 - Flexible

- Safe
- Provide clear feedback and show available options.

4. Hick's Law (MEMORIZE formula meaning)

Definition:

Time to make a decision **increases** as the number of alternatives **increases**.

Formula:

$$[RT = a + b \log_2(n)]$$

Where:

- **RT** = response time
- **a** = non-decision time (perception etc.)
- **b** $\approx$ 0.155 s (processing time per option for humans)
- **n** = number of equally probable choices

Example from slide:

- It takes 2 s to detect alarm & understand it.
 - 5 equally likely buttons to press.
- $$[RT = 2 + 0.155 \log_2(5) \approx 2.36 \text{ s}]$$

Design implications:

- Reduce number of visible choices.
- Group options into categories.

- Use sensible defaults.
-

5. Fitts' Law (MEMORIZE idea)

Definition:

Time to move to a target depends on **distance** and **size** of the target.

Formula:

$$[MT = a + b \log_2 \left(\frac{d}{s} + 1 \right)]$$

- **MT** = movement time
- **a** = 0.230 s
- **b** = 0.166 s
- **d** = distance to target
- **s** = target size

Example from slide:

- Icon of diameter 1" is 6" away.
[
 $MT \approx 0.230 + 0.166 \log_2(6/1 + 1) \approx 0.7 \text{ s}$
]

Design implications:

- Make important targets **bigger and closer** to where pointer/finger usually is.
 - Edges/corners of screen are effectively bigger targets (mouse stops there).
-

6. Golden Ratio & Fibonacci sequence

Golden Ratio (ϕ):

- Ratio ~ 0.618 : smaller segment / larger segment = larger / whole.
- Can be computed as: $((\sqrt{5} - 1) / 2 \approx 0.618)$.
- $1 + 0.618 = 1.618...$ often called **phi** too.
- Used to create:
 - Golden rectangles
 - Golden ellipses
 - Golden triangles
- Research shows people often prefer rectangles based on this ratio.

Fibonacci Sequence:

- Each number = sum of previous two: 1, 1, 2, 3, 5, 8, 13, ...
- Appears in nature: petals, galaxies, hand bones.
- Ratios of adjacent Fibonacci numbers approximate the **golden ratio**.

Design usage:

- Layouts, grids, scaling elements, sizing columns.
-

7. Rule of Thirds

Definition:

Divide area into 3x3 grid (9 rectangles, 4 intersections). Place the main element near an **intersection** for a pleasing composition.

- Very common in photography, poster design.
- If the primary element is too strong and unbalances the composition:

- It may be better to **center** it.

8. Signal-to-Noise Ratio (MEMORIZE meaning)

Definition:

Ratio of **relevant** information (signal) to **irrelevant** information (noise) in a display.

Aim: **maximize** signal-to-noise.

- Maximize **signal**:
 - Clear writing
 - Appropriate graphs
 - Unambiguous icons/labels
 - Simple design → low cognitive load.
- Minimize **noise**:
 - Remove unnecessary data, graphics, lines, decorative elements.
 - Every extra thing steals attention.

PRACTICE – Week 12 Questions

A. 5 MCQs (with answers & explanations)

MCQ 1

Which of the following best describes **anthropomorphism** in HCI?

- A. Designing systems that can detect human emotions via sensors
- B. Attributing human-like qualities to computers, robots, or objects

- C. Designing interfaces to be emotionally neutral
- D. Using AI to classify facial expressions

Answer: B

Explanation:

Anthropomorphism means giving human-like traits (speech, personality, emotions) to non-human objects like computers or robots.

MCQ 2

According to the three levels of emotional design, which level focuses on **usability and actual use** of the product?

- A. Visceral
- B. Behavioral
- C. Reflective
- D. Affective

Answer: B

Explanation:

Behavioral design is about **use** and corresponds to traditional usability: how well the product works in action.

MCQ 3

Which of the following is **NOT** one of the four characteristics of accessible design mentioned in your slides?

- A. Perceptibility
- B. Operability
- C. Efficiency
- D. Forgiveness

Answer: C

Explanation:

The four characteristics are **Perceptibility, Operability, Simplicity, and Forgiveness**. Efficiency is important generally but is not one of the named four.

MCQ 4

Hick's Law primarily tells us that:

- A. Big targets are faster to hit than small ones
- B. Users take longer to decide when there are more choices
- C. Users are happier when interfaces are colorful
- D. Aesthetically pleasing designs are always more usable

Answer: B

Explanation:

Hick's Law states that decision time increases as the number of alternatives increases; response time is a function of the number of choices.

MCQ 5

Which of these is an example of **persuasive technology for behavior change** from your slides?

- A. Nest thermostat's round screen
- B. Plutchik's wheel of emotions
- C. Nintendo's Pocket Pikachu
- D. Tidy Street's chalk visualization has no effect

Answer: C

Explanation:

Pocket Pikachu is designed to change children's behavior by rewarding physical activity; the pet gets angry if you don't move enough.

B. 5 Short Answer Questions (with model answers)

SAQ 1

Explain the difference between automatic and conscious emotions with an example of each.

Model answer:

Automatic emotions are **fast** reactions that appear and disappear quickly, like a brief flash of anger when an app crashes. Conscious emotions develop more **slowly** and last longer, such as ongoing jealousy or resentment about being treated unfairly.

SAQ 2

What are expressive interfaces and how can they both help and harm usability?

Model answer:

Expressive interfaces use **color, icons, sounds, animations, and overall look-and-feel** to convey emotional states and make interaction more appealing. They can help by providing reassuring, fun feedback that makes users more tolerant of minor problems, but can harm usability if they become **intrusive, distracting, or annoying**, causing frustration.

SAQ 3

Define the 80/20 rule and explain how it guides interface design decisions.

Model answer:

The 80/20 rule says that about **80% of effects** come from **20% of causes**. In interface design, around 80% of usage typically comes from 20% of features. This suggests designers should focus on making that 20% of core features extremely good and consider simplifying, de-emphasizing, or removing the less-used 80% to reduce complexity.

SAQ 4

What is the signal-to-noise ratio in design, and why is a high ratio desirable?

Model answer:

Signal-to-noise ratio is the ratio of **relevant information (signal)** to **irrelevant information (noise)** in a display. A high ratio is desirable because it means important information is clearly communicated with minimal distraction; removing unnecessary graphics, text, and decorations helps users focus on what matters.

SAQ 5

List and briefly describe the three levels of emotional design (visceral, behavioral, reflective).

Model answer:

- **Visceral** – Concerns the immediate look, feel, and sound of a product; first impression and sensory appeal.
- **Behavioral** – Concerns how the product works in use; usability, performance, function.

- **Reflective** – Concerns the long-term meaning and personal value of the product; how it fits with identity and memories.
-

C. Case Study (with full answer)

Scenario

Your team is designing a **mobile energy-saving app** that:

- Shows users their household electricity usage
- Encourages them to reduce consumption
- Uses a friendly virtual character to give feedback

You must:

1. Use **emotional interaction concepts** to describe how the app could encourage behavior change.
 2. Apply **one design rule** (from Accessibility, 80/20, Hick's, Fitts', Golden ratio, etc.) to improve the app's interface.
 3. Explain whether adding a human-like personality to the character is a good idea (pros/cons), using the idea of **anthropomorphism**.
-

Model Answer

1. Emotional interaction & behavior change

The app can use emotional interaction by:

- Providing **expressive feedback**:
 - When usage decreases, the virtual character **smiles**, claps, and shows a happy animation to make the user feel proud.

- When usage increases, the character looks concerned but not harsh, nudging the user gently.
- Using **simple infographics and emoticons** for daily usage:
 - Green smiley when under target, yellow neutral when close, red sad face when over target, similar to energy feedback projects and Tidy Street's simple visuals.
- Sending **persuasive reminders** (nudges):
 - "You're very close to beating last week's low usage, keep going!"
 - This uses behavioral-change techniques like personalized messages and recommendations.

This emotional feedback encourages users to feel **motivated** and **engaged**, making it more likely that they will change their behavior.

2. Applying a design rule

Choose **80/20 rule**:

- Data shows that 80% of users mainly care about:
 - Today's usage vs typical
 - Simple tips to reduce it

So the app should:

- Put **today's usage** and a simple comparison (e.g., "10% less than average") right on the **home screen**.
- Hide detailed graphs and advanced settings under secondary screens.

This reduces clutter and maximizes **signal-to-noise**; core features are easy to see, less critical info is not in the way.

(You could also mention **Accessibility**: using simple text and clear charts so people of different abilities can perceive and operate the app.)

3. Anthropomorphism – pros and cons of human-like character

Using a human-like or pet-like character involves **anthropomorphism**, attributing human traits to software.

Pros:

- Can make the app more **enjoyable and motivating**.
- Users may feel an **emotional attachment**, similar to Pocket Pikachu or virtual pets, making them more likely to care about energy usage.
- Friendly, encouraging feedback (“Great job! You saved energy today!”) can positively impact willingness to continue.

Cons:

- If the character **overdoes** it (too chatty, naggy, like Clippy), users may find it annoying, patronizing, or distracting.
- Some users may find an apologizing or overly emotional character **fake** or insincere.

Conclusion:

Anthropomorphism is useful if the character is **subtle, respectful, and supportive**. It should remain a **companion** to the numeric feedback, not a clown or a nagging parent.